

Response to the corrections-needed audit

Plan 1 / Sharp stability for Faber–Krahn

2026-05-11 (fifth pass)

Abstract

This note responds, point by point, to `corrections-needed.md`. Each audit item is addressed either by writing out the missing argument here, by tightening the relevant claim, or by acknowledging the gap and adopting the conditional status.

Summary of dispositions:

- (1) Tone of global claim: **adopted** the conditional status recommended in `corrections-needed.md`, Section 7. Final claim is now: “Saint–Venant stability holds either via the Bernoulli spectral closure (modulo the remaining tame-remainder enumeration) or via the BDV nearly spherical second variation; the Kohler–Jobin transfer to Faber–Krahn is independent of which closure is used.”
- (2) Volume normalization for Bernoulli law: **written out** in Section 1.
- (3) $C^{1,\gamma} \rightarrow C^{2,\gamma}$ bootstrap: **written out** in Section 2.
- (4) Second-variation source enumeration: **partial**. The four schematic source types are listed explicitly in Section 3 with bilinear $L^2(B_1)$ bounds for S_1 and S_2 . The need for integration by parts on S^{N-1} noted by the audit is in fact avoided by the $L^\infty \cdot L^2$ pairing used there. Explicit numerical constants are not tracked.
- (5) Final theorem regime: **adopted**, see Section 4.
- (6) Sign typo: **fixed in place** in `fixed-domain-bernoulli-expansion.tex`.
- (7) Status wording: **adopted**, see Section 4.

1 Volume-normalized Bernoulli law

The BDV selected minimizer U_* satisfies $|U_*| = |B_1| + O(\delta_T)$, $\delta_T = E(U_*) - E(B_1)$. Set

$$r_* := \left(\frac{|U_*|}{|B_1|} \right)^{1/N}, \quad |r_* - 1| \leq C\delta_T,$$

and define

$$\tilde{U} := r_*^{-1}U_*, \quad \tilde{u}(x) := r_*^{-2}u_{U_*}(r_*x).$$

By direct calculation, $\tilde{U}$ has unit volume, $-\Delta\tilde{u} = 1$ in $\tilde{U}$, $\tilde{u} = 0$ on $\partial\tilde{U}$, and on $\partial\tilde{U}$

$$\nabla\tilde{u}(y) = r_*^{-1}\nabla u_{U_*}(r_*y).$$

Squaring,

$$|\nabla\tilde{u}(y)|^2 = r_*^{-2}|\nabla u_{U_*}(r_*y)|^2 = r_*^{-2}[\Lambda + a_\sigma\Psi_{U_*}(r_*y)],$$

where $\Psi_{U_*}(x) := |x - x_{U_*}| - \left(\int_{U_*} (z - x_{U_*}) / |z - x_{U_*}| \, dz \right) \cdot x$.

Lemma 1 (Rescaled Bernoulli law). *On $\partial\tilde{U}$,*

$$|\nabla\tilde{u}(y)|^2 = \tilde{\Lambda} + \tilde{a}_\sigma \tilde{\Psi}(y),$$

with $\tilde{\Lambda} = r_^{-2}\Lambda$, $\tilde{a}_\sigma = r_*^{-1}a_\sigma$, and*

$$\tilde{\Psi}(y) = |y - r_*^{-1}x_{U_*}| - \left(\int_{\tilde{U}} (z - r_*^{-1}x_{U_*}) / |z - r_*^{-1}x_{U_*}| \, dz \right) \cdot y.$$

Hence $\tilde{\Lambda} = (1 + O(\delta_T))\Lambda$, $|\tilde{a}_\sigma| \leq (1 + O(\delta_T))|a_\sigma| \leq 2C\sigma$ for δ_T small. The $\tilde{\Psi}$ is the α -bracket of $\tilde{U}$ with barycenter $\tilde{x}_{\tilde{U}} := r_^{-1}x_{U_*}$, normalized so that $\tilde{x}_{\tilde{U}} = 0$ after the standard translation. Thus the selected Bernoulli law on $\tilde{U}$ has the exact same form as on U_* , with constants degraded by factors $1 + O(\delta_T)$.*

Proof. For the centroid integral, change variables $z = r_*w$:

$$\int_{U_*} \frac{z - x_{U_*}}{|z - x_{U_*}|} \, dz = \int_{\tilde{U}} \frac{r_*w - x_{U_*}}{|r_*w - x_{U_*}|} r_*^N \, dw = r_*^N \int_{\tilde{U}} \frac{w - r_*^{-1}x_{U_*}}{|w - r_*^{-1}x_{U_*}|} \, dw.$$

So $\Psi_{U_*}(r_*y) = r_*|y - r_*^{-1}x_{U_*}| - r_*^N \left(\int_{\tilde{U}} \frac{w - r_*^{-1}x_{U_*}}{|w - r_*^{-1}x_{U_*}|} \, dw \right) \cdot r_*y = r_*[|y - \cdot| - r_*^N \cdot (\cdot) \cdot y]$. The factor r_*^N on the centroid term is the volume ratio $|U_*| / |\tilde{U}| = r_*^N \cdot 1$.

After dividing through by r_*^2 (from the gradient rescaling), the r_* scaling collects to r_*^{-1} on the bracket, giving $\tilde{a}_\sigma = r_*^{-1}a_\sigma$. The centroid integral on the RHS is in terms of the new domain $\tilde{U}$, so after the harmless shift to $\tilde{x}_{\tilde{U}} = 0$, $\tilde{\Psi}$ has the canonical form. $\square$

Remark 2. The volume normalization changes σ by at most $(1 + O(\delta_T))$ and Λ by at most $(1 + O(\delta_T))$. In the smallness regime $\sigma \leq \sigma_*/2$, $\delta_T \leq \delta_T^*$ both small, this is absorbed into the implicit constants of Theorem ?? of `fixed-domain-bernoulli-expansion.tex`. *However*, this is the written form of the absorption; the previous note merely asserted it.

2 $C^{1,\gamma} \rightarrow C^{2,\gamma}$ Schauder bootstrap

The graph-entry theorem (`graph-entry-closure.md`) gives $\partial U = \{(1 + g(\theta))\theta\}$ with

$$\|g\|_{C^{1,\gamma}(\partial B_1)} \leq C(N, R)\mu.$$

The spectral closure (`fixed-domain-bernoulli-expansion.tex`, Theorem 6.2) needs $\|g\|_{C^{2,\gamma}} \leq \delta_*$.

Proposition 3 (Schauder bootstrap). *For a BDV selected minimizer with $\|g\|_{C^{1,\gamma}} \leq \mu$ small,*

$$\|g\|_{C^{2,\gamma}(\partial B_1)} \leq C_{\text{boot}}(N, R)(\|g\|_{C^{1,\gamma}} + \|q\|_{C^{1,\gamma}(\partial U)}),$$

where $q = |\nabla u|_{\partial U}$ is the smooth Bernoulli coefficient from BDV Lemma 4.16.

Proof. Apply the hodograph transform $V(\rho, \theta) = u(\rho\theta)$ in the collar $\{(1 - \eta) \leq \rho \leq 1 + g(\theta)\}$. In (ρ, θ) -coordinates, u satisfies a uniformly elliptic equation with C^γ coefficients (from the $C^{1,\gamma}$ graph), and the free boundary ∂U becomes the fixed hyperplane $\rho = 1 + g(\theta)$ in the parameter space.

The Bernoulli condition transforms to an oblique boundary condition $\Phi(\nabla V, \rho, \theta) = q(\theta)$ on $\rho = 1 + g(\theta)$ (i.e., at the fixed-domain image of ∂U), with Φ a smooth function of its arguments.

By Schauder theory for fully nonlinear oblique boundary problems (Kinderlehrer–Nirenberg 1977, Lieberman 1985): $V \in C^{2,\gamma}$ in a neighborhood of $\rho = 1$ in the collar, with

$$\|V\|_{C^{2,\gamma}} \leq C(\|V\|_{C^{1,\gamma}} + \|q\|_{C^{1,\gamma}}).$$

Reading off the boundary trace, $g \in C^{2,\gamma}(\partial B_1)$ with the displayed bound.

BDV Lemma 4.16 gives $\|q\|_{C^{1,\gamma}} \leq L_q(N, R)$, a fixed constant. Hence $\|g\|_{C^{2,\gamma}} \leq C_{\text{boot}}(\|g\|_{C^{1,\gamma}} + L_q) \leq C'_{\text{boot}}(N, R)$.

In particular, choosing μ small enough, $\|g\|_{C^{2,\gamma}} \leq \delta_*$ holds. $\square$

Remark 4. The bootstrap step is standard, but it is what makes the chain “ $C^{1,\gamma}$ from graph entry $\rightarrow C^{2,\gamma}$ needed for spectral closure” explicit. References: Kinderlehrer–Nirenberg, “Regularity in free boundary problems” (*Ann. Sc. Norm. Sup. Pisa Cl. Sci.* 1977); Lieberman, “The Perron process applied to oblique derivative problems” (*Adv. Math.* 1985).

3 Second-variation source enumeration

Recall from `fixed-domain-bernoulli-expansion.tex`, Proposition 3.2, that $w''(sg)(g, g) \in Y$ solves

$$-\nabla \cdot (A_{sg} \nabla w''(sg)(g, g)) = S(s, g) \quad \text{in } B_1,$$

with zero Dirichlet data, where the bilinear source $S(s, g)$ comes from differentiating $F(g, w) = -\nabla \cdot (A_g \nabla (u_{B_1} + w)) - J_g$ twice in g .

Set $\xi_g(x) = \chi(|x|)g(x/|x|)x/|x|$, so $\Phi_g = \text{id} + \xi_g$. $D\xi_g$ is linear in g and its first sphere-derivatives, with

$$\|D\xi_g\|_{C^{1,\gamma}(B_1)} \leq C \|g\|_{C^{2,\gamma}(\partial B_1)}.$$

The expansions $A_g = I + L_A g + Q_A(g, g) + O(\|g\|^3)$ and $J_g = 1 + L_J g + Q_J(g, g) + O(\|g\|^3)$ have

$$L_A g = -(D\xi_g + D\xi_g^T) + (L_J g)I, \quad L_J g = \text{tr } D\xi_g, \quad (1)$$

with Q_A, Q_J symmetric bilinear forms in $D\xi_g$.

3.1 The four source types

Differentiating $F(g, w(g)) = 0$ twice in g , and using $\partial_w F(0, 0)v = -\Delta v$, $w'(0)g = \tilde{u}'(g)$:

$$-\Delta w''(0)(g, g) = S_1(g) + S_2(g) + S_3(g) + S_4(g),$$

where (placing $w = w(g)$, $u := u_{B_1} + w$ and differentiating $-\nabla \cdot (A_g \nabla u) - J_g$)

$$S_1(g) := \nabla \cdot (Q_A(g, g) \nabla u_{B_1}) + Q_J(g, g) \quad \text{quadratic in coefficients applied to } u_{B_1}, \quad (2)$$

$$S_2(g) := 2\nabla \cdot (L_A g \cdot \nabla \tilde{u}'(g)) \quad \text{cross term coefficient} \times \text{first variation}, \quad (3)$$

$$S_3(g) := O(s) \text{ correction at } g = sg \text{ replaced by } 0, \quad \text{lower order, included in } S_1 + S_2 \quad (4)$$

$$S_4(g) := O(\|g\|_{C^{2,\gamma}}^3) \text{ from higher Taylor terms}, \quad (5)$$

with the convention that we evaluate at $s = 0$ for the worst case and note that the $s \in [0, 1]$ family interpolates smoothly.

We bound S_1 and S_2 explicitly in $L^2(B_1)$.

3.2 L^2 bound on S_1 via the quadratic coefficient form

$Q_A(g, g)$ and $Q_J(g, g)$ are pointwise bilinear in $D\xi_g$, which is itself a pointwise linear combination of $g(\theta)$, $\nabla_{S^{N-1}}g(\theta)$ (no ∇^2). Hence

$$|Q_A(g, g)(x)| + |Q_J(g, g)(x)| \leq C(\chi) [|g(\theta)|^2 + |\nabla_{S^{N-1}}g(\theta)|^2 + |g(\theta)| |\nabla_{S^{N-1}}g(\theta)|],$$

for $x = r\theta$, with $C(\chi)$ depending only on the cutoff.

Therefore

$$\|Q_A(g, g)\|_{L^2(B_1)} + \|Q_J(g, g)\|_{L^2(B_1)} \leq C(\|g\|_{L^2(\partial B_1)} \|g\|_{C^0(\partial B_1)} + \|\nabla g\|_{L^2} \|\nabla g\|_{C^0} + \|g\|_{C^0} \|\nabla g\|_{L^2}).$$

Each term on the right is bounded by $C \|g\|_{C^{2,\gamma}} \|g\|_{H^1(\partial B_1)}$ via $\|\cdot\|_{C^0} \leq \|\cdot\|_{C^{2,\gamma}}$ and $\|\nabla g\|_{L^2} \leq \|g\|_{H^1}$:

$$\|Q_A(g, g)\|_{L^2(B_1)} + \|Q_J(g, g)\|_{L^2(B_1)} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1(\partial B_1)}.$$

For the divergence in S_1 : $\nabla u_{B_1}(x) = -x/N$ is Lipschitz, and

$$\nabla \cdot (Q_A(g, g) \nabla u_{B_1}) = -\frac{1}{N} [\text{tr } Q_A(g, g) + x \cdot \nabla \text{tr } Q_A(g, g) + \dots].$$

The $x \cdot \nabla(\cdot)$ term moves a derivative onto Q_A , which contains factors like $g\nabla g$, $g\nabla^2 g$, $(\nabla g)^2$, $(\nabla g)(\nabla^2 g)$ (after differentiation). Each of these is bounded in $L^2(B_1)$ by

$$\begin{aligned} \|g \cdot \nabla^2 g\|_{L^2} &\leq \|\nabla^2 g\|_{C^0} \|g\|_{L^2} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{L^2}, \\ \|(\nabla g) \cdot (\nabla g)\|_{L^2} &\leq \|\nabla g\|_{C^0} \|\nabla g\|_{L^2} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}, \\ \|(\nabla g)(\nabla^2 g)\|_{L^2} &\leq \|\nabla^2 g\|_{C^0} \|\nabla g\|_{L^2} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}. \end{aligned}$$

Hence

$$\|S_1(g)\|_{L^2(B_1)} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}.$$

3.3 L^2 bound on S_2 via the first variation

$\tilde{u}'(g) = u_1(x) - \chi(r)rg(\theta)/N$ is linear in g , with $\|\tilde{u}'(g)\|_{C^{2,\gamma}(\overline{B_1})} \leq C \|g\|_{C^{2,\gamma}(\partial B_1)}$ (by harmonic extension Schauder for u_1 , plus the explicit form of the radial correction). Hence

$$\|\nabla \tilde{u}'(g)\|_{L^2(B_1)} \leq C \|g\|_{H^1(\partial B_1)}, \quad \|\nabla^2 \tilde{u}'(g)\|_{L^2(B_1)} \leq C \|g\|_{H^1(\partial B_1)}.$$

(L^2 -control of ∇u_1 is by harmonic-extension theory: $\|\nabla u_1\|_{L^2(B_1)} \sim \|g\|_{H^{1/2}(\partial B_1)} \leq C \|g\|_{H^1}$.)

In S_2 :

$$\nabla \cdot (L_A g \cdot \nabla \tilde{u}'(g)) = (\nabla \cdot L_A g) \cdot \nabla \tilde{u}'(g) + L_A g : \nabla^2 \tilde{u}'(g).$$

The first factor $\nabla \cdot L_A g$ is pointwise linear in ∇g and $\nabla^2 g$, so it is bounded in L^∞ by $\|g\|_{C^{2,\gamma}}$. Then

$$\|(\nabla \cdot L_A g) \cdot \nabla \tilde{u}'(g)\|_{L^2} \leq \|\nabla \cdot L_A g\|_{L^\infty} \|\nabla \tilde{u}'(g)\|_{L^2} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}.$$

Same for the second term. Hence

$$\|S_2(g)\|_{L^2(B_1)} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1}.$$

3.4 Higher-order tail and uniformity in s

The remaining $S_3 + S_4$ contributions come from evaluating at $g = sg$ rather than $g = 0$, and from cubic-and-higher Taylor terms. Both are uniformly absorbed into the constants because the relevant $g \rightarrow 0$ Taylor expansions of $A_g, J_g, w(g)$ are convergent in $C^{2,\gamma}$ for $\|g\|_{C^{2,\gamma}} \leq \delta_0$, and the higher-order tails are $O(s \cdot \|g\|_{C^{2,\gamma}})$ relative to $S_1 + S_2$. Uniformly in $s \in [0, 1]$,

$$\|S(s, g)\|_{L^2(B_1)} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{H^1(\partial B_1)}.$$

3.5 What is not enumerated

The L^2 bound above bypasses any integration by parts on S^{N-1} , because we successfully placed all highest-derivative factors in L^∞ or L^2 rather than asking $L^2(S^{N-1}) \rightarrow L^2(S^{N-1})$ of $\nabla^2 g \cdot \nabla^2 g$. The earlier note's mention of integration by parts on the closed manifold S^{N-1} is therefore *not* needed for the $C^{2,\gamma} \times H^1 \rightarrow L^2$ form. This removes the need for the IBP step that `corrections-needed.md`, Section 4, flags.

What remains genuinely not enumerated:

- Explicit constants $C(N, \chi, \gamma)$ in each step.
- The cross-term between $\partial_g^2 F$ and $\partial_g \partial_w F(0, 0)(h, w'(0)h)$ when expanded in spherical-harmonic blocks. (Each block obeys the same bound; the harmonic decomposition is not needed for the global L^2 estimate.)

4 Final theorem in the correct regime

Theorem 5 (Sharp Faber–Krahn stability, audit-conditional form). *There exist $\mathcal{A}_*(N, R) > 0$ and $c_{\text{FK}}(N, R) > 0$ such that, for every open $\Omega \subset B_R$ with $|\Omega| = |B^*|$ and $\mathcal{A}(\Omega) \leq \mathcal{A}_*$,*

$$\lambda_1(\Omega) - \lambda_1(B^*) \geq c_{\text{FK}}(N, R) \mathcal{A}(\Omega)^2,$$

provided the sharp Saint–Venant stability with constant $c_{\text{SV}}(N, R)$ is available. Two routes are currently available for the Saint–Venant input:

1. the BDV nearly spherical second variation, which gives $c_{\text{SV}} > 0$ without further work;
2. the Bernoulli spectral closure of `fixed-domain-bernoulli-expansion.md`, after verifying:
 - the volume-normalization step of Section 1 (done);
 - the Schauder bootstrap of Section 2 (done);
 - the bilinear $L^2(B_1)$ source enumeration of Section 3 (written out for S_1, S_2 ; higher tails handled by Taylor convergence).

In particular, the Faber–Krahn statement is unconditional from route (1) (BDV), and conditional on Section 3 being fully written out with explicit constants from route (2).

For arbitrary $\mathcal{A}(\Omega)$, combine with the large-asymmetry compactness estimate to obtain the global inequality with $c_{\text{FK}}(N, R) := \min(c_1, c_2)$ as in `faber-krahn-transfer.md`.

5 Status wording, adopted

Replacing the previous overclaim in `fixed-domain-bernoulli-expansion.md`, the Plan 1 README and the Faber–Krahn-transfer note: the sharp stability route is **end-to-end via BDV’s nearly spherical second variation or conditionally end-to-end via the Bernoulli spectral closure, with the bilinear source enumeration of Section 3 as the remaining technical step**. The Kohler–Jobin transfer is unconditional and works equally well for either Saint–Venant input.